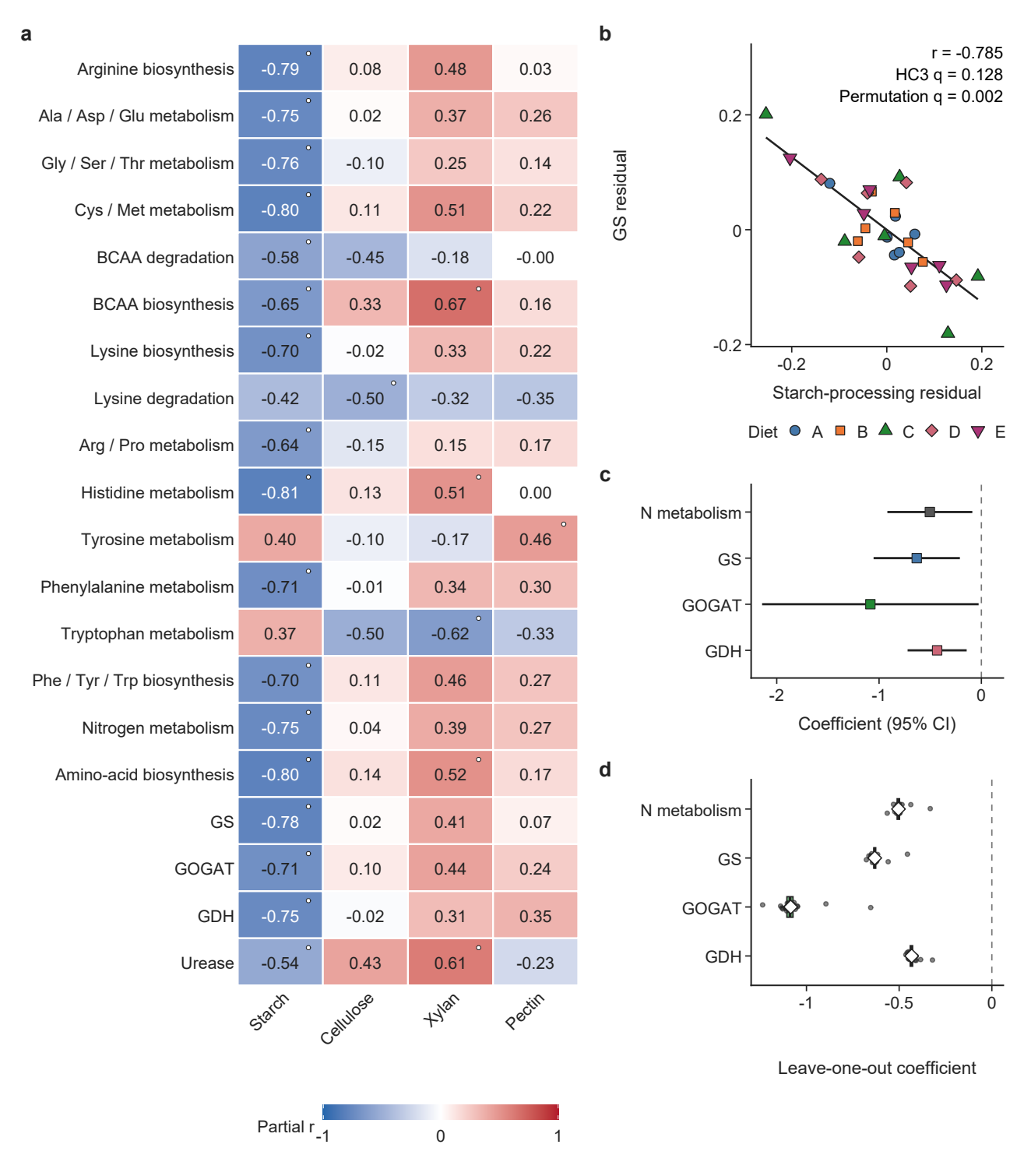
a
Arginine biosynthesis
Ala / Asp / Glu metabolism
Gly / Ser / Thr metabolism
Cys / Met metabolism
BCAA degradation
BCAA biosynthesis
Lysine biosynthesis
Lysine degradation
Arg / Pro metabolism
Histidine metabolism
Tyrosine metabolism
Phenylalanine metabolism
Tryptophan metabolism
Phe / Tyr / Trp biosynthesis
Nitrogen metabolism
Amino-acid biosynthesis
GS
GOGAT
GDH
Urease
-0.79 0.08 0.48 0.03
-0.75 0.02 0.37 0.26
-0.76 -0.10 0.25 0.14
-0.80 0.11 0.51 0.22
-0.58 -0.45 -0.18 -0.00
-0.65 0.33 0.67 0.16
-0.70 -0.02 0.33 0.22
-0.42 -0.50 -0.32 -0.35
-0.64 -0.15 0.15 0.17
-0.81 0.13 0.51 0.00
0.40 -0.10 -0.17 0.46
-0.71 -0.01 0.34 0.30
0.37 -0.50 -0.62 -0.33
-0.70 0.11 0.46 0.27
-0.75 0.04 0.39 0.27
-0.80 0.14 0.52 0.17
-0.78 0.02 0.41 0.07
-0.71 0.10 0.44 0.24
-0.75 -0.02 0.31 0.35
-0.54 0.43 0.61 -0.23
Starch
Cellulose
Xylan
Pectin
Partial r
-1
0
1
b
r = -0.785
HC3 q = 0.128
Permutation q = 0.002
GS residual
0.2
0
-0.2
-0.2
0
0.2
Starch-processing residual
Diet A B C D E
c
N metabolism
GS
GOGAT
GDH
-2
-1
0
Coefficient (95% CI)
d
N metabolism
GS
GOGAT
GDH
-1
-0.5
0
Leave-one-out coefficient